

MADEIRA AP, Portugal

WMO: 085210

Lat: 32.693N

Lon: 16.776W

Elev: 191

StdP: 14.59

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	54.1	55.5	39.1	35.3	63.5	41.5	38.8	60.3	33.6	59.9	29.9	60.5	8.7	350	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	81.8	67.7	79.6	68.4	78.2	68.2	71.9	77.7	70.7	76.7	69.8	75.9	13.3	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.8	110.3	75.9	68.4	105.0	75.1	67.3	101.1	74.5	35.7	77.9	34.6	76.6	33.8	75.9	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.9	25.9	22.9	DB	51.2	87.4	3.0	4.5	49.0	90.7	47.2	93.3	45.6	95.9	43.4	99.2	
			WB	45.6	73.4	2.1	1.1	44.1	74.2	42.8	74.8	41.6	75.4	40.0	76.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.0	61.8	61.2	62.1	63.3	65.6	69.1	72.3	74.0	73.1	70.9	66.8	64.0
	DBStd	5.18	2.57	2.46	2.82	2.56	2.66	2.48	2.61	2.16	2.18	2.49	2.97	2.44
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	464	104	109	99	63	23	1	0	0	0	1	16	48
	CDD50	6223	365	314	374	399	484	574	691	745	694	647	503	433
	CDD65	1211	3	3	8	12	42	125	226	280	244	182	69	17
	CDH74	2472	0	0	5	9	28	135	552	902	608	215	18	0
	CDH80	211	0	0	1	0	1	9	80	70	41	9	0	0
(14) Wind	WSAvg	10.7	10.3	10.9	12.1	10.9	10.3	10.7	12.0	11.5	9.7	9.8	10.0	10.3
(15) Precipitation	PrecAvg	60.0	7.4	7.1	6.0	4.5	2.8	1.5	0.4	0.7	3.5	7.5	8.3	10.3
	PrecMax	89.4	15.5	23.0	12.5	13.0	9.8	5.4	0.8	1.3	11.3	11.5	15.4	20.3
	PrecMin	32.0	2.1	0.3	2.2	1.3	0.6	0.2	0.0	0.2	1.0	2.3	1.9	3.1
	PrecStd	9.9	3.5	4.1	2.3	2.4	1.8	0.9	0.2	0.2	1.9	2.6	2.9	4.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.1	71.0	73.6	74.9	77.2	80.7	87.7	86.4	84.7	80.6	75.8	71.9
		MCWB	62.4	59.4	58.5	60.4	61.9	65.4	64.2	65.2	65.8	68.7	66.8	62.8
	2%	DB	68.3	68.3	70.2	71.0	73.8	77.0	81.2	81.2	80.1	77.6	74.1	70.1
		MCWB	61.6	60.3	60.2	60.0	63.8	66.4	66.2	68.9	69.5	68.0	66.5	62.8
	5%	DB	66.9	66.6	68.1	69.2	71.9	75.2	78.2	79.4	78.4	76.1	72.6	69.0
		MCWB	60.6	59.4	60.0	60.2	63.2	66.2	67.5	69.0	69.0	67.3	65.7	62.4
	10%	DB	65.7	65.2	66.5	67.8	70.2	73.8	76.6	78.1	77.1	74.9	71.4	68.0
		MCWB	59.7	58.4	59.1	59.7	62.4	65.3	67.0	68.6	68.2	67.0	64.8	62.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.9	64.4	64.7	64.7	67.3	70.6	71.0	73.3	73.6	72.5	70.1	66.9
		MCDB	67.9	66.9	68.9	69.7	73.1	76.1	78.1	80.2	79.2	76.7	73.1	69.2
	2%	WB	63.1	62.3	62.9	63.2	65.7	68.8	69.8	71.7	72.0	70.8	68.5	65.2
		MCDB	67.0	66.5	67.3	68.1	71.3	74.8	76.8	78.5	77.6	75.0	72.4	68.5
	5%	WB	61.8	60.8	61.5	62.0	64.5	67.4	68.8	70.4	70.5	69.5	67.1	63.8
		MCDB	66.1	65.4	66.3	67.3	70.3	73.6	76.0	77.4	76.4	74.1	71.6	67.9
	10%	WB	60.5	59.4	60.1	60.7	63.3	66.2	67.9	69.4	69.3	68.4	65.6	62.6
		MCDB	65.1	64.1	65.4	66.5	69.3	72.6	75.2	76.6	75.5	73.5	70.5	67.0
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	6.7	7.3	7.7	7.9	8.0	7.9	7.8	7.6	7.3	6.7	6.4
		MCDBR	7.9	9.0	9.6	9.6	9.8	9.9	10.0	9.5	9.4	8.8	8.3	7.9
		MCWBR	6.1	6.9	7.1	6.7	6.3	6.3	6.4	6.0	6.2	6.0	6.1	6.3
	5% WB	MCDBR	7.4	8.0	8.6	8.6	9.2	9.1	8.9	8.8	8.5	7.6	7.4	6.8
		MCWBR	6.6	6.9	7.1	6.9	6.8	6.6	6.3	6.3	6.5	6.3	6.4	6.5
(40) Clear-Sky Solar Irradiance	taub	0.341	0.362	0.390	0.380	0.383	0.382	0.391	0.395	0.391	0.376	0.367	0.348	(40)
	taud	2.549	2.470	2.383	2.432	2.446	2.472	2.449	2.460	2.487	2.523	2.506	2.542	(41)
	Ebn at Noon	279	282	282	289	287	285	282	281	278	275	266	269	(42)
	Edh at Noon	26	31	37	36	36	35	36	35	33	30	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg	899	1159	1553	1912	2089	2233	2225	2081	1752	1335	956	807	(44)
(45)	RadStd	44	80	79	69	133	110	96	76	57	58	57	41	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page